

Tushar Kumar

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Summary

Results-oriented Software Developer with a passion for creating innovative solutions, cloud-hosted infrastructure, and real-time systems using JavaScript, React, Node.js, and SQL databases. Proven track record in startup environments creating intuitive user dashboards, optimizing data workflows, and deploying resilient applications on AWS (EC2). Driven to apply analytical skills to develop efficient, data-driven applications

Technical Skills

Languages: C, JavaScript, Python, Go

Frameworks & Libs: React.js, Redux / Context API, Node.js, Express.js, TypeScript, Jest/Unit Testing, Tailwind CSS

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2), Docker, Kubernetes, GitHub Actions, PM2, Git, Linux

Core Concepts: REST APIs, Database Design, System Monitoring

Professional Experience

Software Developer Intern, TruesTech IT Solution Pvt Ltd. – Gurugram, Haryana

12/2025 – Present

Technologies: Node.js, Express, MySQL, MongoDB, Redis, Docker, PM2, AWS, Webhooks, Cryptography (AES-256 / RSA), Telegram Bot API

- **Engineered High-Throughput Payment Engine:** Architected core payment processing pipelines handling **1,000+ daily payin/payout transactions**, integrating multi-rail payment gateways with idempotent webhook listeners to ensure zero double-spend or transaction dropping during peak traffic spikes.
- **Designed Hybrid Persistence Architecture:** Structured ACID-compliant **MySQL schemas** with strict isolation levels and row-level locking for wallet ledger balances, paired with **MongoDB** for asynchronous event-driven audit logging, enabling low-overhead query traces across financial workflows.
- **Hardened Multi-Layer Security & Cryptography:** Implemented fine-grained OTP authentication with rate-limiting and encrypted sensitive payloads end-to-end using **AES-256/RSA cryptography** across all REST endpoints to mitigate MITM attacks and comply with financial security standards.
- **Built Dynamic Role-Based Multi-Dashboard Suite:** Developed RBAC-enforced administrative and end-user dashboards featuring contextual UI layouts, dynamic permission gating, and granular transaction access controls across distinct user tiers.
- **Automated Financial Reporting & Reconciliation Engine:** Built an asynchronous batch reporting worker generating downloadable, audit-ready financial summaries (PDF/CSV/Excel) for admins and account-level reconciliation statements for end-users without blocking primary API threads.
- **Deployed AI-Driven Proactive Telegram Alerting:** Integrated a real-time Telegram monitoring bot backed by automated error pattern detection to flag unauthorized access attempts, transaction failures, and server anomalies, reducing Mean Time to Resolution (MTTR) to **<5 minutes**.
- **Optimized Infrastructure & Process Load Balancing:** Deployed node microservices via **PM2 Cluster Mode** on cloud instances, utilizing IPC process coordination and memory management to achieve continuous sub-20ms API response latency and maximum CPU core utilization.

Web Developer Intern, AirFleet Managers – Gurugram, Haryana

02/2025 – 08/2025

Technologies: React, MongoDB, REST APIs, Node.js, Express, Tailwind CSS, EC2, PM2, Docker

- **Designed an Aviation Asset Management Portal** with granular Role-Based Access Control (RBAC), enforcing multi-tenant data isolation and dynamic dashboard permission models across Airlines, Lessors, and MROs.
- **Engineered a low-latency messaging infrastructure** supporting direct and group channels for instant cross-organization communications, eliminating reliance on external messaging tools.

- **Implemented automated report generation engine** capable of rendering production-ready, stakeholder-grade PDFs, PPTs, and Excel sheets with embedded analytical charts, optimizing rendering pipelines to keep file sizes under **10MB** even for multi-page datasets.
- **Developed interactive real-time data visualizers and dynamic Gantt charts** to track project milestones, utilizing live data feeds and client-side role filtering to restrict internal progress views to staff while keeping external stakeholder dashboards clean.
- **Containerized Redis and core microservices via Docker**, standardizing multi-environment CI/CD deployments and trimming base image sizes down to **<250MB**.
- **Configured PM2 cluster mode across multi-core AWS EC2 instances**, establishing request load balancing and process management to maintain sub-15ms response times and high system availability.

Web Developer Intern, Sanrachna Foundation – South West Delhi, Delhi

06/2024 – 07/2024

Technologies: React, MongoDB, REST APIs, Node.js, Express, Tailwind CSS, WordPress

- **Redesigned a publication portal** that boosted user engagement by **30%**, reaching over **10,000 monthly active users**.
- **Automated content distribution pipelines** by implementing dynamic categorization and integrated newsletter workflows, reducing manual efforts by **50%**.
- **Developed a centralized planning tool**, reducing editorial turnaround by **35%**.
- **Created a gamified web experience** that transformed abstract psychological topics into accessible visual narratives, raising engagement duration by **40%**

Projects

NEXUS RTC - Cross-Platform Enterprise Desktop Collaboration Suite

Technologies: TypeScript, React, Electron, Node.js, Express, WebSockets, WebRTC, Docker, Tailwind CSS

- **Engineered a cross-platform desktop application** using **Electron**, **React**, and **TypeScript**, delivering low-latency peer-to-peer real-time video, audio, and desktop screen sharing.
- **Developed a high-performance signaling server** using **Node.js**, **Express**, and **WebSockets**, optimizing WebRTC communication by handling **10,000+ daily SDP and ICE candidate transmissions**.
- **Integrated native OS screen-sharing capabilities** using Electron's desktopCapturer API and IPC contextBridge communication for dynamic source selection with real-time frame previews.
- **Built real-time media controls** enabling dynamic track enabling/disabling for microphone muting and camera toggles without tearing down peer connections.
- **Containerized backend services** with multi-stage **Docker** builds on Linux base images, ensuring predictable deployment environments and seamless network port exposure.

Education

SGT University, Gurugram, HR – Master of Computer Applications (MCA)

08/2025 – Present

(Available for full-time job)

SGT University, Gurugram, HR – Bachelor of Computer Applications (BCA)

08/2022 – 08/2025

Languages

- English
- Hindi